PHY202 Practice Exam on chapter(s): 16, 17

1. $\beta = 61.9\,\text{dB} \because v_w = \left(331\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\,\text{K}}}, I = \frac{(\Delta p)^2}{2\rho v_w}, \beta(\text{dB}) = 10\log_{10}\left(\frac{I}{I_0}\right), I_0 = 1\times10^{-12}\frac{\text{W}}{\text{m}^2}$

2. $v_{max} = 0.0737\,\text{m/s} \because v(t) = \sqrt{\frac{k}{m}}X\sin\frac{2\pi t}{T}$ and the max of sine's range is 1.

3. $f = 1.49\,\text{s} \because T_{SHO} = 2\pi\sqrt{\frac{m}{k}}$

4. $f = 4.55\times10^3\,\text{Hz} \because f_n = n\frac{v_w}{4L}, v_w = \left(331\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\,\text{K}}}$

5. $k = 9.60\times10^4\frac{\text{N}}{\text{m}} \because \vec{F} = -k\Delta\vec{x}.\ k = -\frac{\vec{F}}{\Delta\vec{x}}$

6. $PE = 1.58\,\text{J} \because PE = \frac{1}{2}kx^2$

7. $f = 3.70\times10^6\,\text{Hz} \because T = \frac{1}{f}$

8. $d = 5.24\times10^3\,\text{m} \because \vec{v}_{avg} = \frac{\Delta\vec{x}}{\Delta t}$

9. $L = 0.2858\,\text{m} \because f_n = n\frac{v_w}{4L}, v_w = \left(331\frac{\text{m}}{\text{s}}\right)\sqrt{\frac{T}{273\,\text{K}}}$

10. $L = 6.97\,\text{cm} \because T_{pend} = 2\pi\sqrt{\frac{L}{g}}$

11. $a = 0.773\,\% \because Z = \rho v, a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$

12. $d = 1.26\,\text{m} \because \Delta E_t = W_{nc} = -f_k d, f_k = \mu_k N = \mu_k mg, PE = \frac{1}{2}kx^2$

13. $f = 221.0\,\text{Hz} \because f_{obs} = f_s\left(\frac{v_w}{v_w \pm v_s}\right)$ (review the Doppler effect)